

ARPAN HAIT

Full Stack Developer | MERN Stack Specialist

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[LinkedIn](#) | [GitHub](#) | [Codolio](#)

Professional Summary

Adaptable Software Developer with hands-on experience building responsive web applications and scalable API integrations using **React**, **Next.js**, **TypeScript**, and **Python (FastAPI/Django)**. Proficient in state management, asynchronous workflows, and **UI performance optimization**. Strong foundation in modern development workflows including **Git**, **RESTful APIs**, and **cloud** deployments.

Technical Skills

Frontend & Core Web: React.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Bootstrap

API & Integrations: RESTful APIs, Axios, JWT, OAuth, State Management

Development Tools: Git/GitHub, Firebase, Netlify, Vercel, Postman, AWS

Backend Knowledge: Node.js, Express.js, Python, Django, FastAPI, MongoDB

Professional Experience

Generative AI Image Annotator

Jan 2026 – Apr 2026

RT Network Solutions Pvt. Ltd.

WB, Kolkata

- Annotated **185+** high-resolution images weekly for Generative AI model training while following strict annotation guidelines.
- Audited over **2,075** visual datasets, contributing to a **97%** project accuracy rate.
- Improved annotation efficiency by approximately **15%** through optimized contextual tagging and quality assurance practices.

Projects

Krishi Mitra – Multilingual Voice AI Agent [Repository](#) Tech Stack: Next.js 14, React, Python, TypeScript, SQLite, WebRTC

- Architected a production-grade **real-time multilingual Voice AI platform** using **LiveKit Agents**, **Deepgram Nova-3 STT**, **Murf Falcon TTS**, and **Gemini 3.1**, enabling voice interaction across **3 Indic languages** with **100%** strict script-matching.
- Reduced AI response latency to **sub-second speeds** by engineering a **6-turn context-window pruning engine** and configuring LLM with **thinking_budget=0**, eliminating **504 DEADLINE_EXCEEDED** timeouts.
- Engineered a context-preserving **2-way multi-agent handoff protocol** between the primary agricultural advisor and plant pathology specialist, enabling seamless session delegation without requiring users to repeat conversational context.
- Built an asynchronous, non-blocking background engine with **<2ms response time** for **Twilio outbound price-alert calls**, **SMTP/IMAP government-email synchronization**, and a **Next.js 14 Control Center** with **120 FPS GPU-accelerated UI components** and real-time **SSE telemetry streams**.

Industrial AI Anomaly Detection System [Repository](#)

Tech Stack: Next.js, React, TypeScript, FastAPI, Scikit-Learn

- Developed a real-time industrial anomaly detection platform using Next.js and FastAPI, integrating **Isolation Forest** and **Random Forest** models to analyze multivariate sensor telemetry for predictive maintenance.
- Reduced system monitoring time by **40%** for operators by replacing static logs with an interactive Next.js dashboard and live **Web-Socket** telemetry streaming.

Deepfake Detection Framework [Repository](#)

Tech Stack: React, FastAPI, Python, MTCNN, EfficientNet-B0

- Built a responsive React frontend integrated with a Python FastAPI backend running **MTCNN** and **EfficientNet-B0** across **5** asynchronous REST API endpoints for real-time inference.
- Sped up validation workflows by **30%** by designing an async React media upload pipeline that processes multi-format inference metrics without UI blocking.

Education

Narula Institute of Technology

2024 – 2028

Maulana Abul Kalam Azad University of Technology (MAKAUT)

WB, Kolkata-700109

Bachelor of Technology (B.Tech) in Computer Science & Engineering (AI/ML)

SGPA: **8.8/10**

Certifications

- Google** – Build a Website on Google Cloud [LINK](#) | **Cisco** – Cyber Smart AI Prime [LINK](#)

Coding Profiles

Codolio :- [LeetCode](#) | [CodeChef](#) | [HackerRank](#)